Week 2 – SOC Internship Report

1. Introduction

In modern cybersecurity environments, Security Operations Centers (SOCs) play a critical role in detecting, analyzing, and responding to security incidents. With the increasing number of cyber threats such as ransomware, phishing attacks, and data breaches, it is essential for SOC analysts to effectively prioritize alerts, classify incidents, and follow structured response procedures.

This report focuses on three key theoretical areas: Alert Priority Levels, Incident Classification, and Basic Incident Response. Understanding these concepts helps in improving the efficiency of security monitoring and ensures timely and effective handling of security incidents.

2. Theoretical Knowledge

2.1 Alert Priority Levels:

SOC environment mein bahut saare alerts generate hote hain, isliye unhe priority ke basis par categorize karna zaroori hota hai. Alerts ko generally four levels mein divide kiya jata hai: Critical, High, Medium aur Low.

Critical alerts wo hote hain jisme immediate action required hota hai jaise ransomware attack ya data breach. High priority alerts serious hote hain jaise unauthorized admin access. Medium alerts suspicious activity show karte hain jaise multiple failed login attempts, aur Low alerts informational hote hain jaise port scanning.

Alert priority decide karne ke liye kuch important factors consider kiye jate hain jaise asset criticality (production server ya test system), exploit likelihood (public exploit available hai ya nahi), aur business impact (financial ya operational loss).

Example ke liye Log4Shell vulnerability ka CVSS score 9.8 hai, isliye ise Critical category mein rakha jata hai.

CVSS (Common Vulnerability Scoring System) ek standard scoring system hai jo vulnerabilities ki severity measure karta hai. Iske alawa SOC tools jaise Splunk risk scoring use karke alerts ko prioritize karte hain.

2.2 Incident Classification:

Incident classification ka matlab hai security incidents ko unke type ke basis par categorize karna. Isse investigation easy ho jati hai aur response fast hota hai.

Common incident types mein malware attacks, phishing attacks, DDoS attacks, insider threats aur data exfiltration include hote hain.

Example: Agar koi employee unauthorized data export karta hai to use insider threat category mein classify kiya jata hai.

Standard frameworks jaise MITRE ATT&CK, ENISA taxonomy aur VERIS use kiye jate hain incidents ko properly label karne ke liye. MITRE ATT&CK attacker techniques ko map karta hai jaise phishing ke liye T1566.

Incident ko aur better samajhne ke liye contextual metadata add kiya jata hai jaise affected system, timestamp, source IP aur Indicators of Compromise (IOCs) jaise malicious file hash ya URL.

2.3 Basic Incident Response:

Incident response ek structured process hai jisme security incidents ko detect, contain aur resolve kiya jata hai.

Iska lifecycle 6 phases mein hota hai: Preparation, Identification, Containment, Eradication, Recovery aur Lessons Learned.

Preparation phase mein tools aur playbooks ready kiye jate hain. Identification mein incident detect kiya jata hai. Containment mein system isolate kiya jata hai. Eradication mein malware remove kiya jata hai. Recovery mein systems restore kiye jate hain aur last mein lessons learned phase mein improvements identify kiye jate hain.

Incident response ke dauraan kuch important procedures follow kiye jate hain jaise system isolation, evidence preservation (memory dump, logs, hashing), proper communication aur automation tools (SOAR) ka use.

3. Practical Knowledge

3.1 Alert Management Practice

1. **Alert Classification System**

An alert classification system was created using Google Sheets to organize and categorize security alerts. Each alert was mapped based on type, priority, and MITRE ATT&CK technique. For example, phishing attacks were mapped to T1566, while ransomware was mapped to T1486. This structured approach helps in efficient alert handling and faster decision-making in SOC environments.

2. Alert Prioritization Using CVSS

Alerts were prioritized using CVSS (Common Vulnerability Scoring System) scores to determine severity levels. High-impact threats such as ransomware were assigned a CVSS score of 9.8 (Critical), while low-risk activities like port scanning were assigned lower scores (3.5). This prioritization ensures that critical threats are handled immediately.

A	B	C	D	E	F
Alert ID	**Type**	**Priority**	**MITRE Technique**	**Description**	**CVSS Score**
1	Phishing	High	T1566	Suspicious email with malicious link	8
2	Ransomware	Critical	T1486	File encryption detected	9.8
3	Port Scan	Low	T1046	Multiple ports scanned	3.5
4	Brute Force	Medium	T1110	Multiple failed login attempts	6.5
5	Phishing Email	High	T1566	Suspicious link received from unknown sender	8

Figure 1: CVSS Table

3. Dashboard Visualization

A dashboard was created using Google Sheets to visualize alert distribution based on severity levels. A pie chart was used to represent the proportion of Critical, High, Medium, and Low alerts. This helps SOC analysts quickly identify high-risk alerts and focus on urgent threats.

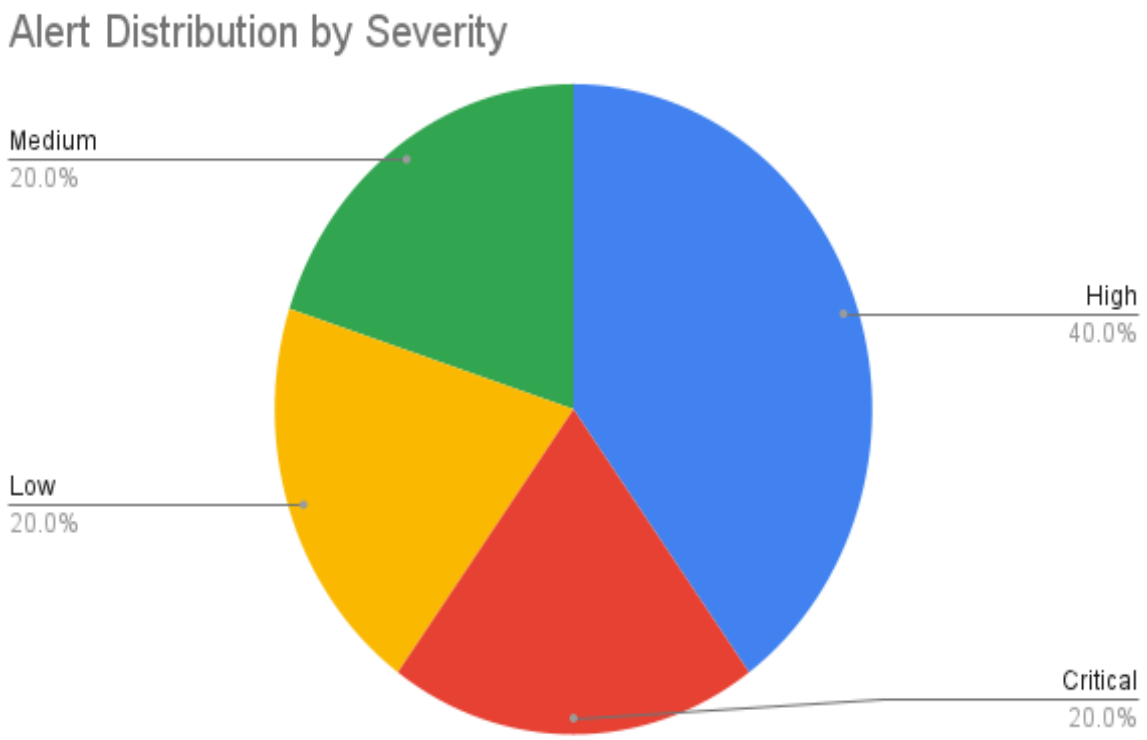

Figure 2: Pie Chart Visualization

4. Incident Ticket Creation

An incident ticket was created to document a ransomware attack scenario. The ticket included key details such as title, description, indicators of compromise (file and IP), priority level, and assigned analyst. This demonstrates how incidents are tracked and managed in real-world SOC operations.

Title: [Critical] Ransomware Detected on Server-X

Description:
Indicators of Compromise (IOCs):
- File: crypto_locker.exe
- IP Address: 192.168.1.50

The system has shown ransomware behavior including file encryption and abnormal activity. Immediate containment and investigation are required to prevent further spread.

Priority: Critical
Assignee: SOC Analyst
Status: Open

Figure 3: Incident Creation

5. **Escalation Procedure**

An escalation email was drafted to notify the Tier 2 team about a critical ransomware incident. The email included important details such as affected system, attack indicators, and urgency of response. This ensures proper communication and timely resolution of high-priority incidents.

Subject: Escalation: Critical Ransomware Incident on Server-X

Dear Tier 2 Team,

A critical ransomware incident has been identified on Server-X. The system is showing active encryption behavior linked to the file crypto_locker.exe. The attack appears to originate from IP address 192.168.1.50.

Initial containment measures have been taken, but further investigation is required. Immediate escalation is necessary to prevent further damage.

Please prioritize this case and initiate advanced response actions.

Regards,
SOC Analyst

Figure 4: Escalation Email

3.2 Response Documentation

1. **Incident Response Report**

A structured incident response report was created following standard formats. It included:

- Executive Summary
- Timeline of events
- Impact analysis
- Remediation steps
- Lessons learned

This documentation helps in understanding the incident lifecycle and improving future response strategies.

1. Executive Summary
A phishing email containing a suspicious link was identified and analyzed. The email posed a risk of credential theft and unauthorized access. Immediate action was taken to isolate the threat and prevent user compromise.

2. Timeline
- 14:00: Suspicious email detected
- 14:10: Alert classified as phishing (T1566)
- 14:20: User account monitored
- 14:30: Malicious link blocked

3. Impact Analysis
The attack had the potential to compromise user credentials and sensitive data. No confirmed data breach occurred due to timely response.

4. Remediation Steps
- Blocked malicious URL
- Educated user about phishing risks
- Updated email filters

5. Lessons Learned
User awareness is critical in preventing phishing attacks. Early detection and response helped avoid damage.

Figure 5: Incident Report

2. Investigation Log

An investigation log was maintained to record all actions taken during the incident. Each step was documented with timestamps, ensuring traceability and accountability during the investigation process.

Timestamp	Action
2025-08-18 14:00:00	Alert detected
2025-08-18 14:10:00	Classified as phishing
2025-08-18 14:20:00	User activity monitored
2025-08-18 14:30:00	Malicious link blocked

Figure 6: Investigation Log

3. Phishing Checklist

A phishing investigation checklist was created to standardize the analysis process. It included steps such as analyzing email headers, verifying sender details, checking links via VirusTotal, and identifying affected users. This ensures consistency in handling phishing incidents.

Phishing Investigation Checklist

1. Analyze email headers
2. Verify sender email address
3. Check link reputation using VirusTotal
4. Identify affected users
5. Block malicious links
6. Educate user

Figure 7: Checklist

4. Post-Mortem Analysis

A post-mortem analysis was conducted to evaluate the incident response. It highlighted the importance of early detection, user awareness, and improved security controls. Recommendations were provided to enhance future security posture.

The incident highlighted the importance of early detection and user awareness. Implementing stronger email filtering and regular security training can reduce phishing risks. Proper documentation and quick response ensured no data loss occurred. Future improvements include automated threat detection and enhanced monitoring.

Figure 8: Post Mortem

3.3 Alert Triage Practice

1. Alert Simulation

A mock alert representing brute-force SSH activity was created. The alert included details such as source IP, description, priority, and status. This simulates real-world alert triage scenarios in SOC environments.

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH	192.168.1.100	Medium	Open

Figure 9: Alert Table

2. Threat Intelligence Validation

The source IP (192.168.1.100) was analyzed using VirusTotal. The IP was identified as a private address with no malicious detections (0/94). This indicates that the activity is likely internal or a false positive. Continuous monitoring was recommended.

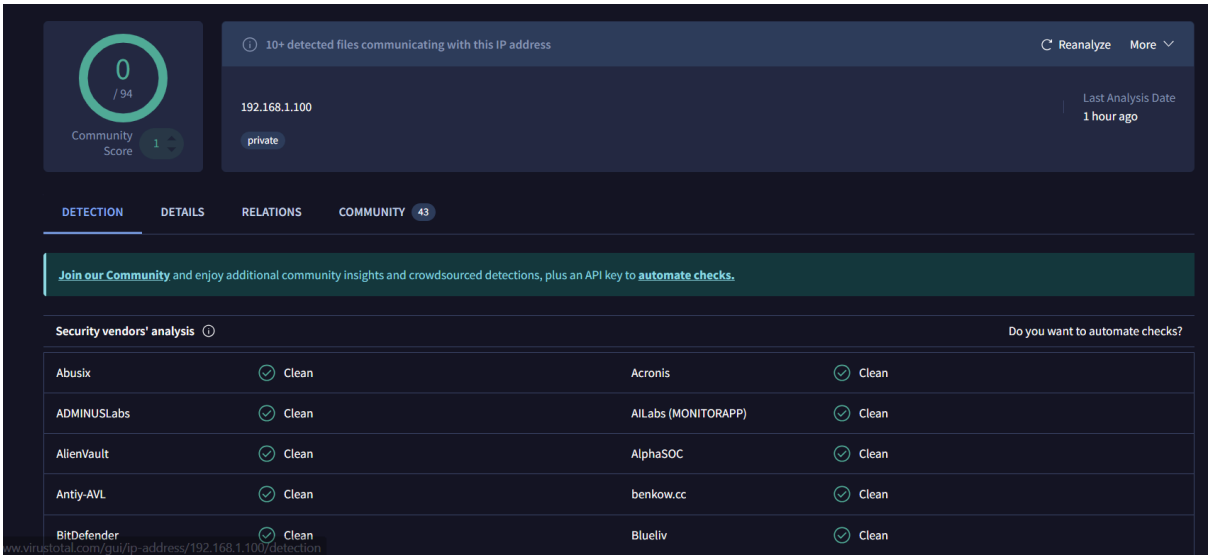

Figure 10: VirusTotal

3. Triage Analysis

The alert was analyzed and marked as "Under Analysis." Based on findings, it was identified as a likely false positive due to the use of a private IP address. This demonstrates the importance of validating alerts before escalation.

Alert ID	Description	Source IP	Priority	Status	Verdict
002	Brute-force SSH	192.168.1.100	Medium	Open	Likely False Positive (Internal IP)

Figure 11: Triage Table

3.4 Evidence Preservation

1. Volatile Data Collection

Volatile data was collected using the netstat -ano command to identify active network connections and processes. This helps detect suspicious activity and ongoing connections during an incident.

2. Memory Dump Collection

A simulated memory dump was created to represent volatile memory acquisition. This step is essential for analyzing running processes and detecting malware present in system memory.

```
Simulated Memory Dump Collection
System: Windows Machine
Tool: Velociraptor (Simulated)

Memory data was collected to analyze running processes and detect malicious activity.
```

Figure 12: Memory Dump Collection

3. Hash Generation

A SHA256 hash was generated for the memory dump file using the certutil command. Hashing ensures the integrity of collected evidence and prevents tampering.

```
PS C:\Users\arinv\OneDrive\Desktop\TXT FILES> certutil -hashfile memory_dump.txt SHA256
SHA256 hash of memory_dump.txt:
a98efce253a31bcd26b9769fbb9796b13716c4de38c74b8462a11a394b208e94
CertUtil: -hashfile command completed successfully.
PS C:\Users\arinv\OneDrive\Desktop\TXT FILES>
```

Figure 13: Hash Generated

4. Evidence Documentation

An evidence table was created to maintain chain-of-custody. It included details such as item name, description, collection date, and hash value. This ensures proper handling and documentation of digital evidence.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Dump	SOC Analyst	2026-03-26	a98efce253a31bcd26b9769fbb9796b13716c4de38c74b8462a11a394b208e94

Figure 14: Evidence Table

3.5 Capstone Project: Full Alert-to-Response Cycle

1. Attack Simulation

A simulated attack scenario was created representing exploitation of a vulnerable service. The attack originated from a private IP address and targeted a system to demonstrate SOC workflow.

Attack Simulation:
A simulated attack was performed representing a VSFTPD backdoor exploit. The attack originated from IP address 192.168.1.100 targeting a vulnerable server.

Figure 15: Attack Simulation

2. **Detection and Analysis**

The attack was detected and mapped to MITRE ATT&CK technique T1190 (Exploitation of Public-Facing Application). The alert was analyzed and validated using threat intelligence tools.

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-26 11:00:00	192.168.1.100	Simulated exploit attack	T1190

Figure 16: Detection & Triage

3. **Response Actions**

The affected system was isolated, and the suspicious IP address was blocked. Network activity was monitored to ensure no further malicious actions occurred.

4. **Final Report Summary**

The incident was handled successfully through timely detection, analysis, and response. No data loss occurred, and preventive measures were implemented to avoid future incidents.

5. **Recommendations**

- Regular system updates and patching
- Implementation of intrusion detection systems
- User awareness training
- Continuous monitoring of network activity

4. Conclusion

This practical implementation demonstrated the complete SOC workflow, including alert classification, triage, incident response, and evidence preservation. The use of tools like Google Sheets, VirusTotal, and command-line utilities helped simulate real-world security operations effectively. The project enhanced understanding of SOC processes and incident handling techniques.